

Problem

A series RLC circuit is described by

$$L \frac{d^2 i(t)}{dt^2} + R \frac{di(t)}{dt} + \frac{i(t)}{C} = 15$$

Find the response when $L = 0.5 \text{ H}$, $R = 4 \Omega$, and $C = 0.2 \text{ F}$. Let $i(0^-) = 7.5 \text{ A}$ and $[di(0^-)/dt] = 0$.

Step-by-step solution

Step 1 of 4

Consider the differential equation which describes the step responses of series RLC .

$$L \frac{d^2 i(t)}{dt^2} + R \frac{di(t)}{dt} + \frac{i(t)}{C} = 15$$

Substitute 0.5 H for L , 4Ω for R , and 0.2 F for C .

$$0.5 \frac{d^2 i(t)}{dt^2} + 4 \frac{di(t)}{dt} + \frac{i(t)}{0.2} = 15$$

$$\frac{d^2 i(t)}{dt^2} + 8 \frac{di(t)}{dt} + 10i(t) = 30$$

Consider the following series RLC circuit initial conditions.

$$i(0^-) = i(0^+) = i(0)$$

$$\frac{di(0^-)}{dt} = \frac{di(0^+)}{dt} = \frac{di(0)}{dt}$$

Step 2 of 4

Apply the Laplace transform to the differential equation.

$$\left(s^2 I(s) - si(0^-) - \frac{di(0^-)}{dt} \right) + 8(sI(s) - i(0^-)) + 10I(s) = \frac{30}{s}$$

Substitute 7.5 A for $i(0^-)$ and 0 for $\frac{di(0^-)}{dt}$.

$$\left(s^2 I(s) - 7.5s - \frac{di(0^-)}{dt} \right) + 8(sI(s) - 7.5) + 10I(s) = \frac{30}{s}$$

$$I(s)(s^2 + 8s + 10) - (7.5s + 60) = \frac{30}{s}$$

$$I(s)(s^2 + 8s + 10) = \frac{30}{s} + 7.5s + 60$$

$$I(s) = \frac{7.5s^2 + 60s + 30}{s(s^2 + 8s + 10)}$$

$$I(s) = \frac{7.5s^2 + 60s + 30}{s(s + 1.5505)(s + 6.4495)}$$

Step 3 of 4

Decompose $I(s)$ into simple terms by taking the partial fraction.

$$I(s) = \frac{A}{s} + \frac{B}{(s+1.5505)} + \frac{C}{(s+6.4495)}$$

$$\frac{7.5s^2 + 60s + 30}{s(s^2 + 8s + 10)} = \frac{A}{s} + \frac{B}{(s+1.5505)} + \frac{C}{(s+6.4495)}$$

$$7.5s^2 + 60s + 30 = A(s+1.5505)(s+6.4495) + Bs(s+6.4495) + Cs(s+1.5505) \dots (1)$$

Substitute 0 for s in equation (1).

$$30 = A(6.4495)(1.5505)$$

$$A = \frac{30}{(6.4495)(1.5505)}$$

$$= 3$$

Substitute -1.5505 for s in equation (1).

$$-45 = B(-1.5505)(4.899)$$

$$B = \frac{-45}{(-1.5505)(4.899)}$$

$$B = 5.924$$

Substitute -6.4495 for s in equation (1).

$$-45 = C(-6.4495)(-4.899)$$

$$C = \frac{-45}{(-6.4495)(-4.899)}$$

$$= -1.424$$

Substitute obtained A, B, and C values in equation $I(s)$.

$$\frac{7.5s^2 + 60s + 30}{s(s^2 + 8s + 10)} = \frac{3}{s} + \frac{5.924}{(s+1.5505)} - \frac{1.424}{(s+6.4495)}$$

$$I(s) = \frac{3}{s} + \frac{5.924}{(s+1.5505)} - \frac{1.424}{(s+6.4495)}$$

Write the Laplace transform pairs those are used to find $i(t)$.

$$u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s}$$

$$e^{-at}u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s+a}$$

Apply inverse Laplace transform on both sides of equation $I(s)$.

$$i(t) = 3u(t) + 5.924e^{-1.5505t}u(t) - 1.424e^{-6.4495t}u(t)$$

$$i(t) = \left[3 + 5.924e^{-1.5505t} - 1.424e^{-6.4495t} \right] u(t)$$

Step 4 of 4

Therefore, the response $i(t)$ of an RLC circuit is

$$\boxed{\left[3000 + 5924e^{-1.5505t} - 1424e^{-6.4495t} \right] u(t) \text{ mA}}.$$